

*Over fit: Train error low – Test error High*

*Underfit: Train error high*

*Normal fit: Train error low – Test error low*

### **Bias – Variance Trade off**

*It is a curve shows about bias and Variance*

*X – axis: Model complexity*

*For example we have 100 input columns*

$$\text{Model – 1: } y = b_0 + b_1 * x_1$$

$$\text{Model – 2: } y = b_0 + b_1 * x_1 + b_2 * x_2 + b_3 * x_3$$

$$\text{Model – 3: } y = b_0 + b_1 * x_1 + b_2 * x_2 + \dots + b_{10} * x_{10}$$

$$\text{Model – 4: } y = b_0 + b_1 * x_1 + b_2 * x_2 + \dots + b_{50} * x_{50}$$

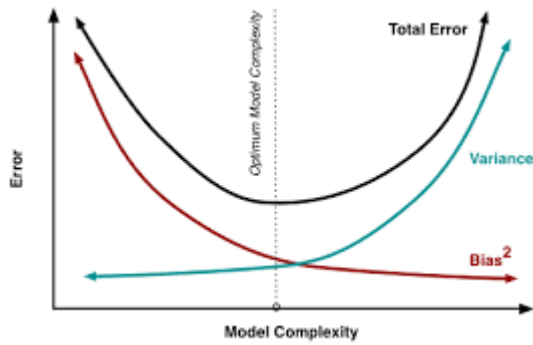
*Model complexity will increase , when we add the more features (input columns)*

*Bias: Bias is also an error generally (actual – predictions)*

- *Bias is high : for less features (model complexity is less)*
- *Bias is low: for maximum features (Model complexity is high)*

*Variance:*

- *Variance is high : for maximum features (model complexity is high)*
- *Variance is low: for less features (Model complexity is low)*



$$\text{Total error} = \text{Bias}^2 + \text{variance}$$

*Bias is related to Train error*

*Variance is related to Test error*

*Underfit: Train error High =====> High Bias – Low variance*

*normal fit: Train error low – Test error low =====> Low Bias – Low variance*

*Overfit: Train error low – Test error is High =====> Low Bias – High variance*

## Two questions

*When model will be overfit?*

- *Overfitting is happend because we are using more features*
- *More features ===== More dimensions*
- *Training data is very high*
- *Over training (Deep learning will learn this)*
- *if any model gives 100% accuracy means its overfitted model*
- *No model will give 100% accuracy*

*What you will do to avoid the overfit?*

- *Select the important features (feature selection methods)*
  - *How to select the features*
    - *Variance threshold method*
    - *Correlation*
    - *Mutual information*

- $p$  – value
  - Forward selection – Backward selection
  - Chi square
  - Anova
  - L2 Regularization (Ridge)
- Dimension reduction techniques
  - Principle component analysis (PCA)
  - Singular value decomposition (SVD)
- Data augmentation (Deep Learning)
 

*more complexity + less data then will use data augmentation method*



*Upsampling*

*SMOTE*

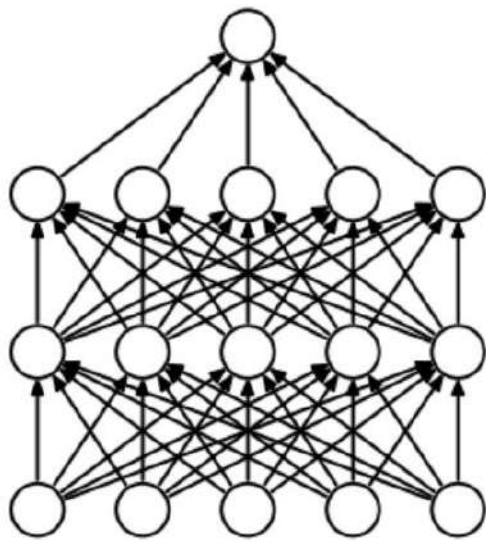
**Related to Complexity decrease methods:**

*Early stoping*

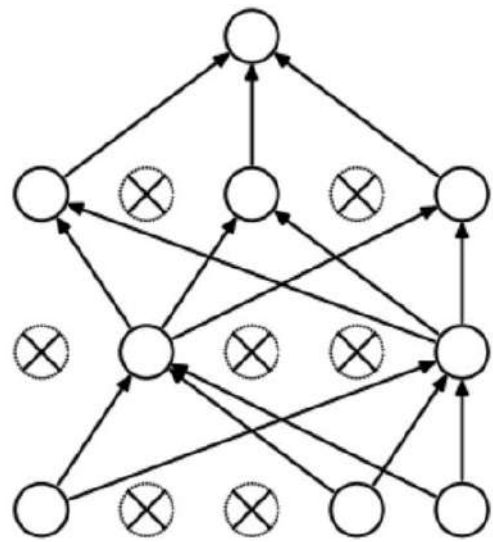
*Tree pruning (Decisio tree)*

*Regularization (lasso and ridge)*

*Drop out (Deep learning will drop some neurons)*



(a) Standard Neural Net



(b) After applying dropout.

$$Total\ error = Bias^2 + variance$$